Appendix: Full Code

Below is the full code used for this project. It is all the code chunks I had in my quarto doc put into a large chunck that doesn't run.

$$
\begin{aligned}
y_t &= \beta_0 + \beta_1 \times \text{population}_t + \beta_2 \times \mathbb{1}_{t \in [2020,2021]}(t) + x_t \\
x_t &\sim \text{ARIMA}(p,d,q) \\
\nabla^d x_t &= \phi_1 x_{t-1} \dots \phi_p x_{t-p} + w_t + \theta_1 w_{t-1} + \theta_q w_{t-q} \\
t &= 1950, \dots, 2023 \quad w_t \sim \text{WN}(0, \sigma_w^2)
\end{aligned}
$$

```r
library(tidyverse)
library(astsa)
library(forecast)
library(ggplot2)
library(gridExtra)
library(modelsummary)

owid_co2_clean <- read_csv(here::here("data/cleaned/owid_co2_clean.csv"))
co2_ts <- ts(owid_co2_clean[201:274,]$co2, start = 1950) # 1950 to 2023
population <- ts(owid_co2_clean[201:274,]$population, start = 1950) # 1950 to 2023


par(mfrow=c(1,2))

plot(co2_ts,
     main = "Annual Total Global Carbon Dioxide Emmissions",
     xlab = "Year",
     ylab = "Metric tonnes of Carbon Dioxide")
plot(population,
     main = "Annual Total Global Population",
     xlab = "Year",
```

```r
     ylab = "Billions of People"
     )



year1 <- 2007
year2 <- 2014
year3 <- 2021

full_years <- time(co2_ts)
train_end_year <- 2021
forecast_start_year <- 2022
forecast_end_year <- 2023
h_forecast <- forecast_end_year - forecast_start_year + 1
covid_dummy_full <- numeric(length(full_years))
covid_dummy_full[full_years == 2020] <- 1
covid_dummy_full[full_years == 2021] <- 1
full_xreg_matrix <- cbind(
  population = as.numeric(population), # make sure is full series
  covid = covid_dummy_full
)
xreg_train <- as.matrix(full_xreg_matrix[1:72]) # for ARIMA part
covid_dummy_train_ts <- covid_dummy_full[1:72]

co2_train1 <- window(co2_ts, 1950, year1)
co2_train2 <- window(co2_ts, 1950, year2)
co2_train3 <- window(co2_ts, 1950, year3)

pop_train1 <- window(population, 1950, year1)
pop_train2 <- window(population, 1950, year2)
pop_train3 <- window(population, 1950, year3)

fit1 <- lm(log(co2_train1) ~ pop_train1)
fit2 <- lm(log(co2_train2) ~ pop_train2)
fit3 <- lm(log(co2_train3) ~ pop_train3 + covid_dummy_train_ts)

fitted_values1 <- ts(as.numeric(fit1$fitted.values), start = 1950)
fitted_values2 <- ts(as.numeric(fit2$fitted.values), start = 1950)
fitted_values3 <- ts(as.numeric(fit3$fitted.values), start = 1950)

par(mfrow=c(2,3))
```

```r
plot(log(co2_train1),
     main = "1950-2007",
     xlab = "Year",
     ylab = "Log CO2 (tonnes)")
lines(fitted_values1, col = "red")

plot(log(co2_train2),
     main = "1950-2014",
     xlab = "Year",
     ylab = "Log CO2 (tonnes)")
lines(fitted_values2, col = "red")

plot(log(co2_train3),
     main = "1950-2022",
     xlab = "Year",
     ylab = "Log CO2 (tonnes)")
lines(fitted_values3, col = "red")

plot(resid(fit1))
plot(resid(fit2))
plot(resid(fit3))
```

```r
par(mfrow=c(1,3))
plot(diff(resid(fit3), differences = 2),
     main = "Plot of Residuals of Log(CO2). d = 2")
invisible(acf(diff(resid(fit3), differences = 2)))
invisible(acf(diff(resid(fit3), differences = 2), type = "partial"))
```

```r
par(mfrow=c(3,2))
model1 <- Arima(log(co2_train1), order = c(0, 2, 2), xreg = pop_train1, include.mean = FALSE
# Plot the fitted values
plot(log(co2_train1), lwd = 2,
     main = "Fitted Regression with ARIMA(0,2,2) Errors Model",
     ylab = "log(CO2) (tonnes)",
     xlab = "1950-2007")
lines(model1$fitted, col = "red", lwd = 2)
```

```
lines(fitted_values1, col = "blue", lty = 2, lwd = 1)
h <- 16
last_year <- end(co2_train1)[1] # 2007
future_time_vals <- (last_year + 1):(last_year + h)
forecast_population_vals <- owid_co2_clean[259:274,]$population
co2_forecast1 <- forecast(model1, xreg = forecast_population_vals, h = h)

# Plot the forecast
actual_co2 <- owid_co2_clean[259:274,]$co2
years <- c(2008:2023)
plot(co2_forecast1,
     main = "Forecasted Values for 2008-2023",
     xlim = c(2000, 2023))
points(x = years, y = log(actual_co2), col = "orange", pch = 19)


model2 <- Arima(log(co2_train2), order = c(0, 2, 2), xreg = pop_train2, include.mean = FALSE
# Plot the fitted values
plot(log(co2_train2), lwd = 2,
     main = "Fitted Regression with ARIMA(0,2,2) Errors Model",
     ylab = "log(CO2) (tonnes)",
     xlab = "1950-2014")
lines(model2$fitted, col = "red", lwd = 2)
lines(fitted_values2, col = "blue", lty = 2, lwd = 1)
h <- 9
last_year <- end(co2_train1)[1] # 2014
future_time_vals <- (last_year + 1):(last_year + h)
forecast_population_vals <- owid_co2_clean[266:274,]$population
co2_forecast2 <- forecast(model2, xreg = forecast_population_vals, h = h)

# Plot the forecast
actual_co2 <- owid_co2_clean[266:274,]$co2
years <- c(2015:2023)
plot(co2_forecast2,
     main = "Forecasted Values for 2015-2023",
     xlim = c(2000, 2023))
points(x = years, y = log(actual_co2), col = "orange", pch = 19)


model3 <- Arima(log(co2_train3), order = c(0,2,1), xreg = xreg_train)
# Plot the fitted values
plot(log(co2_train3), lwd = 2,
```

```r
     main = "Fitted Regression with ARIMA(0,2,1) Errors Model",
     ylab = "log(CO2) (tonnes)",
     xlab = "1950-2021")
lines(model3$fitted, col = "red", lwd = 2)
lines(fitted_values3, col = "blue", lty = 2, lwd = 1)
future_years <- forecast_start_year:forecast_end_year
future_time_vals <- as.numeric(future_years)
forecast_xreg_matrix <- as.matrix(full_xreg_matrix[73:74])
co2_forecast3 <- forecast(model3, xreg = forecast_xreg_matrix, h = h_forecast)

# Plot the forecast
actual_co2 <- log(owid_co2_clean[273:274,]$co2)
years <- forecast_start_year:forecast_end_year
plot(co2_forecast3,
     main = "Forecasted Values for 2022 and 2023",
     xlim = c(2000, 2023))
points(x = years, y = actual_co2, col = "orange", pch = 1)
```

```r
par(mfrow=c(2,1))

forecast_2022_data <- data.frame(
  Model = c("Model 1", "Model 2", "Model 3"),
  Lower_95 = exp(c(co2_forecast1$lower[15, 2], co2_forecast2$lower[8, 2], co2_forecast3$lowe
  Upper_95 = exp(c(co2_forecast1$upper[15, 2], co2_forecast2$upper[8, 2], co2_forecast3$uppe
  Forecast_2022 = exp(c(co2_forecast1$mean[15], co2_forecast2$mean[8], co2_forecast3$mean[1]
  Observed_2022 = owid_co2_clean$co2[273]
)
forecast_2022_data$Error <- forecast_2022_data$Observed - forecast_2022_data$Forecast

forecast_2023_data <- data.frame(
  Model = c("Model 1", "Model 2", "Model 3"),
  Lower_95 = exp(c(co2_forecast1$lower[16, 2], co2_forecast2$lower[9, 2], co2_forecast3$lowe
  Upper_95 = exp(c(co2_forecast1$upper[16, 2], co2_forecast2$upper[9, 2], co2_forecast3$uppe
  Forecast_2023 = exp(c(co2_forecast1$mean[16], co2_forecast2$mean[9], co2_forecast3$mean[2]
  Observed_2023 = owid_co2_clean$co2[274]
)
forecast_2023_data$Error <- forecast_2023_data$Observed - forecast_2023_data$Forecast
```

```r
kableExtra::kable(forecast_2022_data, digits = 3, caption = "The detailed forecast statistics
kableExtra::kable(forecast_2023_data, digits = 3, caption = "The detailed forecast statistics
```

```r
fit4 <- lm(co2_train2 ~ pop_train2)
modelsummary::modelsummary(models = list("Fit from 2007" = fit2, "Fit from 2014" = fit4))
```

```r
invisible(acf2(diff(resid(fit1), differences = 2), max.lag = 50))

invisible(acf2(diff(resid(fit2), differences = 2), max.lag = 50))

invisible(acf2(diff(resid(fit3), differences = 2), max.lag = 50))
```

```r
model4 <- sarima(log(co2_train1), 0, 2, 2, xreg = as.matrix(pop_train1))
model5 <- sarima(log(co2_train2), 0, 2, 2, xreg = as.matrix(pop_train2))
model6 <- sarima(log(co2_train3), 0, 2, 1, xreg = xreg_train)
```

```r
kableExtra::kable(as.data.frame(model4$ttable))
```

```r
kableExtra::kable(as.data.frame(model5$ttable))
```

```r
kableExtra::kable(as.data.frame(model6$ttable))
```